Frustums

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

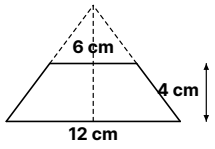
KEY WORDS frustum · cone *(koeko)* · similar triangles *(tapatoru rite)* · cross-section *(hautanga whakapae)*

A The full cone ●●○

Remember: A frustum is a cone with its top cut off. The cut-off part is a smaller, similar cone. The cross-section shows two similar triangles: use them to find the full height H.

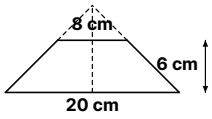
1. Each cross-section shows a frustum (solid lines) and the cut-off cone (dashed). The diameters and the height of the frustum are given. Find the full height H of the cone.

a)



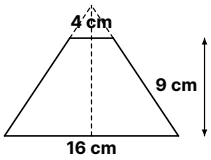
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b)



_____

c)



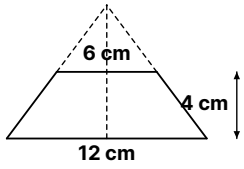
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B Volume of a frustum of a cone ●●●

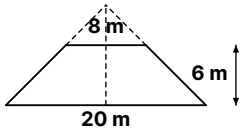
Remember: Volume of the frustum = volume of the full cone − volume of the cut-off cone. Find H first. Give answers to 1 decimal place.

2. Find the volume of each frustum.

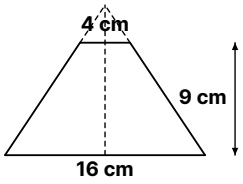
a)



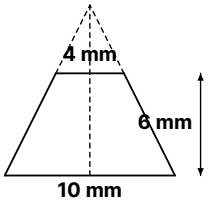
b)



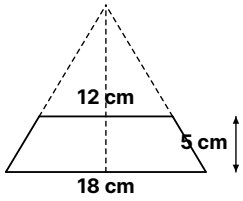
c)



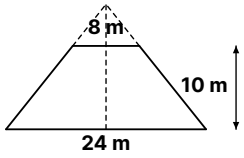
d)



e)



f)

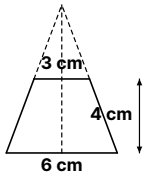


C Frustums of pyramids ●●●

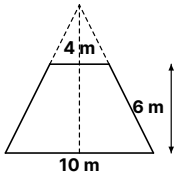
Remember: The same idea works for a pyramid: the cut-off top is a smaller, similar pyramid. Use the base edges instead of radii in the similar triangles.

3. Each solid is a square-based pyramid with its top cut off, shown in cross-section. Find its volume.

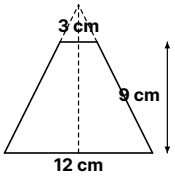
a)



b)

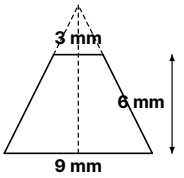


c)

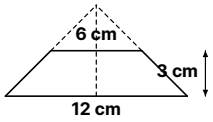


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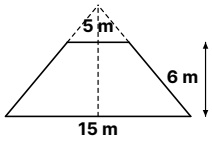
d)



e)



f)



D Problems ●●●

4. A flower pot is a frustum of a cone with a top radius of 10 cm, a bottom radius of 6 cm and a height of 12 cm. Find its volume, to the nearest cm³.

5. A bucket is a frustum of a cone: 14 cm radius at the top, 10 cm radius at the bottom and 25 cm high. How many litres does it hold, to 1 decimal place? (1 L = 1000 cm³.)

6. A plant pot is 15 cm tall with a top radius of 12 cm and a bottom radius of 8 cm. Find the volume of potting mix it holds, to the nearest cm³.

7. A lampshade is a frustum of a cone with a bottom diameter of 20 cm, a top diameter of 10 cm and a height of 24 cm. Explain how you would find the area of fabric needed for its curved surface, without working it out.

★ Extension

8. A frustum of a cone has radii 6 cm and 3 cm and a height of 4 cm. Its curved surface area is $\pi(R + r)s$, where s is the slant height of the frustum. Find the curved surface area, to 1 decimal place.

9. A drinking glass is a frustum with a top radius of 4 cm, a bottom radius of 3 cm and a height of 12 cm. Use the formula $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ to find how many millilitres it holds, to the nearest mL (1 mL = 1 cm³).

10. A frustum has radii R and r and height h. Show that its volume is $\frac{1}{3}\pi h(R^2 + Rr + r^2)$.

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Frustums

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

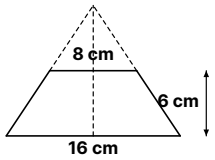
KEY WORDS frustum · cone *(koeko)* · similar triangles *(tapatoru rite)* · cross-section *(hautanga whakapae)*

A The full cone ●●○

Remember: A frustum is a cone with its top cut off. The cut-off part is a smaller, similar cone. The cross-section shows two similar triangles: use them to find the full height H.

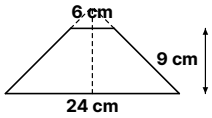
1. Each cross-section shows a frustum (solid lines) and the cut-off cone (dashed). The diameters and the height of the frustum are given. Find the full height H of the cone.

a)



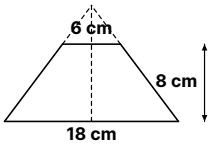
_____

b)



_____

c)



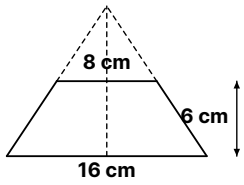
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B Volume of a frustum of a cone ●●●

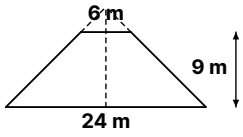
Remember: Volume of the frustum = volume of the full cone − volume of the cut-off cone. Find H first. Give answers to 1 decimal place.

2. Find the volume of each frustum.

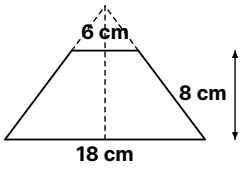
a)



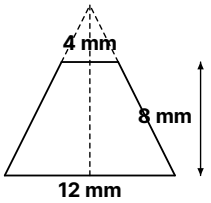
b)



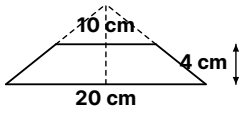
c)



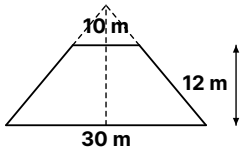
d)



e)



f)

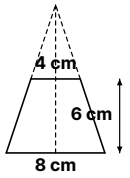


C Frustums of pyramids ●●●

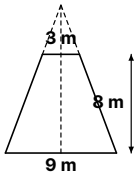
Remember: The same idea works for a pyramid: the cut-off top is a smaller, similar pyramid. Use the base edges instead of radii in the similar triangles.

3. Each solid is a square-based pyramid with its top cut off, shown in cross-section. Find its volume.

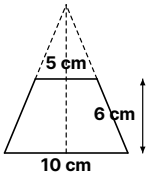
a)



b)

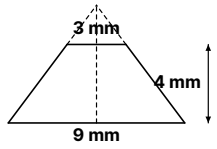


c)

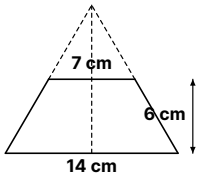


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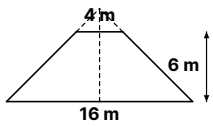
d)



e)



f)



D Problems ●●●

4. A flower pot is a frustum of a cone with a top radius of 12 cm, a bottom radius of 8 cm and a height of 10 cm. Find its volume, to the nearest cm³.

5. A bucket is a frustum of a cone: 15 cm radius at the top, 10 cm radius at the bottom and 30 cm high. How many litres does it hold, to 1 decimal place? (1 L = 1000 cm³.)

6. A plant pot is 18 cm tall with a top radius of 16 cm and a bottom radius of 10 cm. Find the volume of potting mix it holds, to the nearest cm³.

7. A lampshade is a frustum of a cone with a bottom diameter of 24 cm, a top diameter of 12 cm and a height of 30 cm. Explain how you would find the area of fabric needed for its curved surface, without working it out.

★ Extension

8. A frustum of a cone has radii 8 cm and 4 cm and a height of 6 cm. Its curved surface area is $\pi(R + r)s$, where s is the slant height of the frustum. Find the curved surface area, to 1 decimal place.

9. A drinking glass is a frustum with a top radius of 5 cm, a bottom radius of 4 cm and a height of 12 cm. Use the formula $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ to find how many millilitres it holds, to the nearest mL (1 mL = 1 cm³).

10. A frustum has radii R and r and height h. Show that its volume is $\frac{1}{3}\pi h(R^2 + Rr + r^2)$.

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